

3(a)(i)	The <u>electric force per unit positive charge</u> placed at that point.	B1
(a)(ii)	Electric <u>field strength</u> at a point is <u>equal to the negative potential gradient</u> at that point {Accept use of eqn, $E = -\frac{dV}{dx}$, as symbols are already defined in qn; cannot use 'r' in place of 'x'}	B1
(a)(iii)	From the graph, at $x = 15.0$ cm, the <u>gradient is positive</u> . The <u>electric field strength is thus negative</u> , which means that it is <u>pointing leftwards/towards sphere A</u> .	B1
(b)	At $x = 15.0$ cm, the potential due to both spheres is zero. $V_A + V_B = 0$ $\frac{Q_A}{4\pi\epsilon_0 x} + \frac{Q_B}{4\pi\epsilon_0 (d - x)} = 0$ $\frac{Q_A}{0.15} = \frac{-0.48 \times 10^{-9}}{(0.60 - 0.15)}$ $Q_A = -1.6 \times 10^{-10} \text{ C}$	C1 C1 A1
(c)	loss in electric potential energy = gain in kinetic energy $U_i - U_f = K.E._f - K.E._i$ $-1.60 \times 10^{-19} \times (0 - 140) = \frac{1}{2} \times 9.11 \times 10^{-31} \times \left[v_f^2 - (4.0 \times 10^6)^2 \right]$ $v_f \approx 8.1 \times 10^6 \text{ m s}^{-1}$	C1 C1 A1

$1/(e)i$	Resistance is the ratio of the potential difference across the component to the current	R1
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